Brief Solutions

[2pnts.] 1. Formulate the following as a Linear programming problem:
Raw materials M_1, M_2, M_3 is used for the manufacture of two products P_1, P_2. The problem is to decide on the amount of P_1, P_2 to be manufactured to maximize profit. The raw material requirement and the profit associated with each unit of P_1, P_2 is given below.

	Raw material required	(per unit of P_1, P_2)		Profit per unit
	M_1	M_2	M_3	
P_1	4	3	2	Rs 5
P_2	2	1	3	Rs 7
Raw material available	15	20	12	

Answer

Solution: Let x_1, x_2 be the number of units of P_1, P_2 produced, respectively. Then the LPP formulation is:
Max $5x_1 + 7x_2$
subject to
$4x_1 + 2x_2 \leq 15$
$3x_1 + x_2 \leq 20$
$2x_1 + 3x_2 \leq 12$
$x_1, x_2 \geq 0$

2. Consider the following LPP **(P)**:

$$\begin{aligned} \text{Min } & -2x_1 + 3x_2 \\ \text{subject to } & x_1 + x_2 \geq 1 \\ & x_1 \leq 2 \\ & -x_1 + x_2 \geq 1 \\ & x_1 \geq 0, x_2 \geq 0. \end{aligned}$$

[1pnts.] (a) Draw a rough sketch of Fea**(P)**.

[2pnts.] (b) Find all the extreme points and the extreme directions of Fea**(P)**.

[2pnts.] (c) If possible find an optimal solution and the optimal value of **(P)**.

Answer

Solution:

(a)

(b) The extreme points are : $[0,1]^T$ and $[2,3]^T$.
The extreme directions are: Any positive scalar multiple of $[0,1]^T$.

(c) **(P)** has an optimal solution since it is minimization problem and $\mathbf{c}^T\mathbf{d} = 3\beta > 0$ for all extreme directions $\mathbf{d} = \beta[0,1]^T$, $\beta > 0$. By evaluating the value of the objective function at each of the extreme points we get that the unique optimal solution is $[0,1]^T$ and the optimal value is 3.

3. Consider the following LPP **(P)**:

$$\text{Min } c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4$$

$$\text{subject to} \quad x_1 - 2x_2 + 3x_3 - x_4 \leq -1$$
$$2x_1 + x_2 + x_3 \leq 6$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0.$$

Let $\mathbf{x}_0 = [1, 1, 0, 0]^T$.

[2pnts.] (a) If possible find a $\mathbf{d}$ such that $\mathbf{x}_0 \pm \alpha\mathbf{d} \in Fea(\mathbf{P})$ for all $\alpha > 0$, sufficiently small.

[3pnts.] (b) If possible find an extreme point of Fea$(\mathbf{P})$ which lies on all the **defining hyperplanes** of Fea$(\mathbf{P})$ on which $\mathbf{x}_0$ lies.

[3pnts.] (c) If $c_1 = -3$ and $c_4 = 1$ then does $(\mathbf{P})$ necessarily have an optimal solution? Justify with proper arguments.

Solution:

(a) $\mathbf{x}_0 = [1, 1, 0, 0]^T$ lies on three LI defining hyperplanes of Fea(P), $x_3 = 0, x_4 = 0$ and $x_1 - 2x_2 + 3x_3 - x_4 = -1$. We have to pick a $\mathbf{d}$ which is orthogonal to the normals of those three LI defining hyperplanes, which are given by, $[0, 0, 1, 0]^T, [0, 0, 0, 1]^T, [1, -2, 3, -1]^T$. The only possible such $\mathbf{d}$'s are $\gamma[2, 1, 0, 0]^T$, $\gamma \neq 0$.
No marks awarded if you have **only** given $\mathbf{d} = 0$.

(b) From the constructive proof of finding an extreme point we get that atleast one of $\mathbf{x}_0 \pm \alpha\mathbf{d}$ will give an extreme point for some α, since $\mathbf{x}_0$ already lies on three LI defining hyperplanes. The first obstruction as we move along the positive and negative direction of $\mathbf{d} = [2, 1, 0, 0]^T$ starting from $\mathbf{x}_0$ (there will be an obstruction since we cannot move indefinitely on both directions due to non negtaivity constraints) will give a feasible point which will lie on one more LI defining hyperplane, hence will be an extreme point of Fea(P). For $\alpha = \frac{3}{5}$, we get $\mathbf{x}_0 + \alpha\mathbf{d} = [\frac{11}{5}, \frac{8}{5}, 0, 0]^T$ is a required extreme point which lies on the four LI defining hyperplanes, $x_3 = 0, x_4 = 0$, $x_1 - 2x_2 + 3x_3 - x_4 = -1$ and $2x_1 + x_2 + x_3 = 6$.
For $\alpha = \frac{1}{2}$, we get $\mathbf{x}_0 - \alpha\mathbf{d} = [0, \frac{1}{2}, 0, 0]^T$ is a required extreme point which lies on the four LI defining hyperplanes, $x_3 = 0, x_4 = 0$, $x_1 - 2x_2 + 3x_3 - x_4 = -1$ and $x_1 = 0$.
If you have found the extreme points by direct calculations with justifying arguments you will get full credit.

(c) The directions of Fea(P) are defined by the following inequalities:

$$d_1 - 2d_2 + 3d_3 - d_4 \leq 0$$
$$2d_1 + d_2 + d_3 \leq 0$$
$$d_1 \geq 0, d_2 \geq 0, d_3 \geq 0, d_4 \geq 0.$$

Because of the second inequality and the non negativity constraints the only $\mathbf{d}$ which satisfies the above inequalities are given by
$\gamma[0, 0, 0, 1]^T$, $\gamma > 0$. Since $\mathbf{c}^T\mathbf{d} = \gamma > 0$ and $(\mathbf{P})$ is a minimization problem so it has optimal solution.